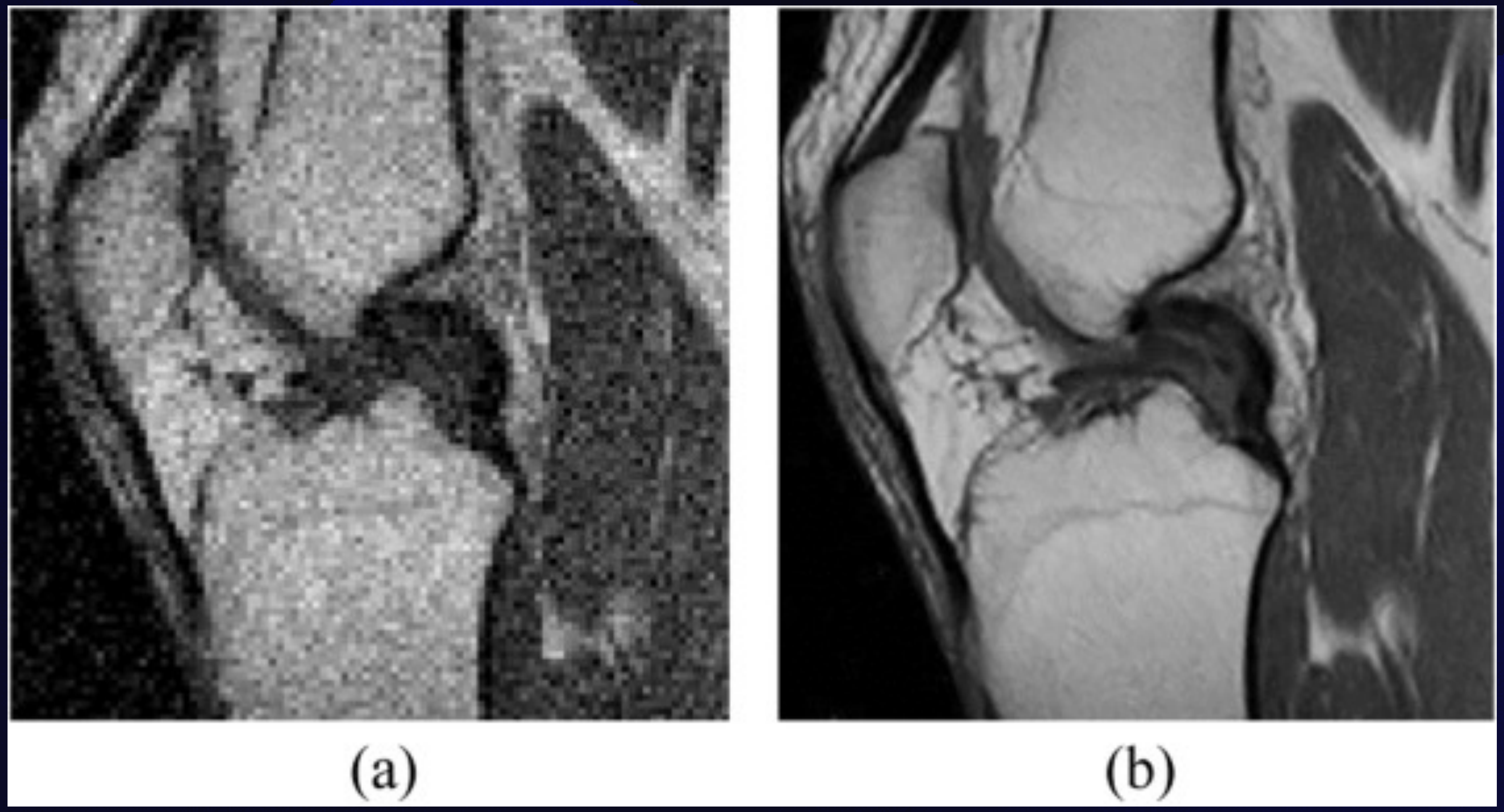
ME228 Course Project

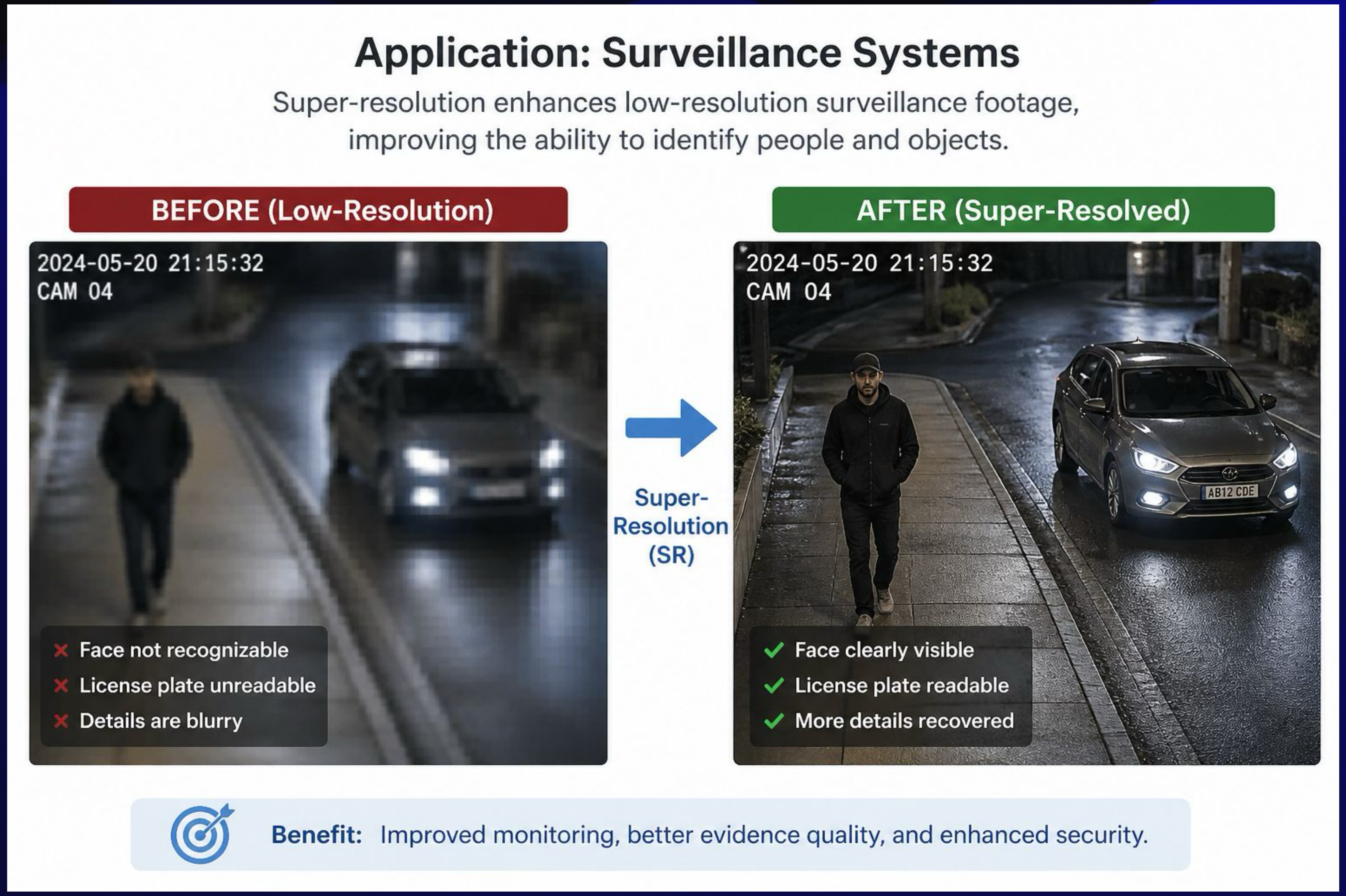
Single Image Super Resoluton

Gaurav Kothari (24b2204), Shaunik Jain (24b2183), Vatsh Van (24b2259), Vihaan Thaker (24b2130)

The Problem - Super Resolution

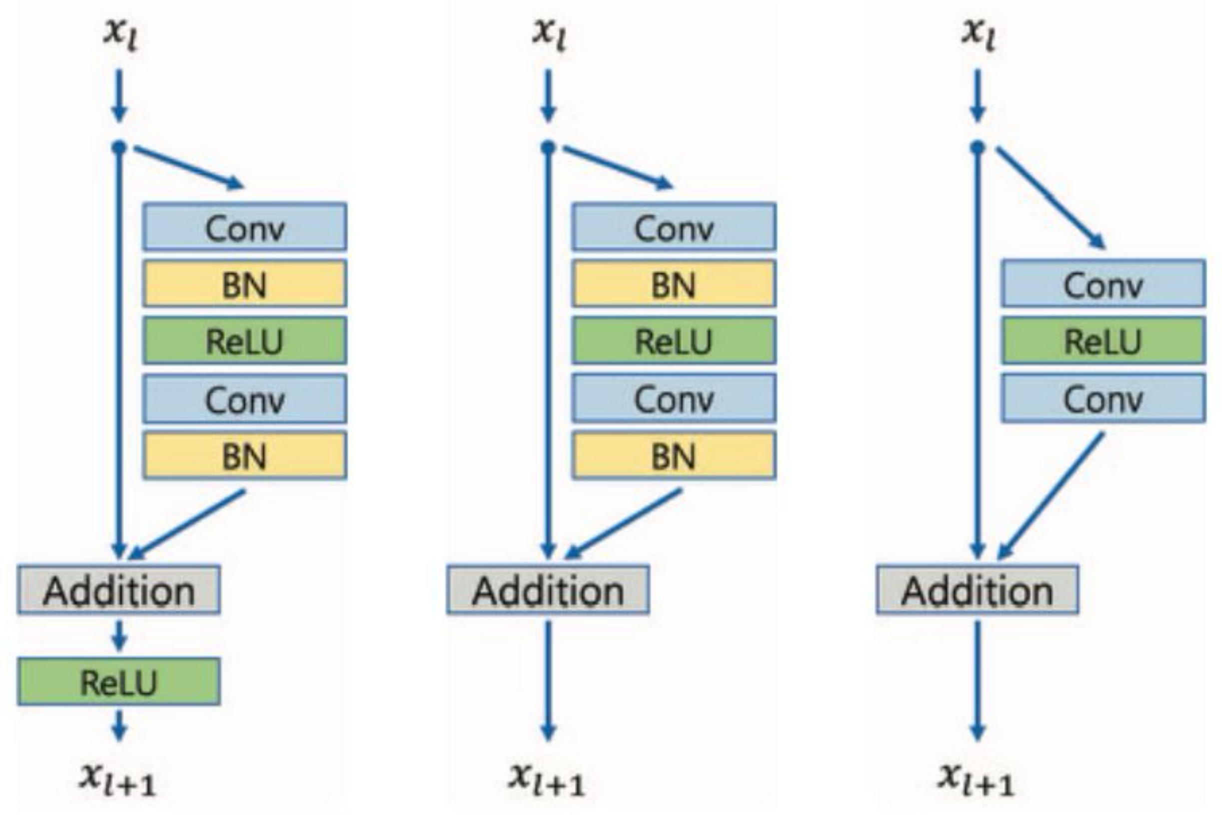
- Super Resolution aims to reconstruct a high-resolution image from a low-resolution input, recovering fine details that are lost during processes like downsampling, noise, or compression. Ideally, there are multiple HR images that can be created from a single LR image. To address this, deep learning models are used to learn mappings from LR to HR images by capturing statistical patterns from large datasets. Super-resolution has important applications in areas such as medical imaging, satellite imagery, surveillance, and consumer photography, where improving image clarity is crucial.





Implemented DL Model (Enhanced Deep Residual Network)

- The EDSR model is a deep convolutional neural network designed specifically for improving image resolution by learning complex mappings from LR to HR images. It is built using a large number (=32 here) of residual blocks, where each block refines feature representations while preserving important information through skip connections. The model operates primarily in the LR space to reduce computational cost and only performs upsampling at the end using sub-pixel convolution, which efficiently increases spatial resolution. By increasing both the depth (no. of blocks) and width (no. of feature channels), and removing unnecessary components like batch normalization, EDSR achieves high-quality reconstruction with sharper details and improved performance compared to earlier methods.



(a) Original (b) SRResNet (c) Proposed

Figure 2: Comparison of residual blocks in original ResNet, SRResNet, and ours.

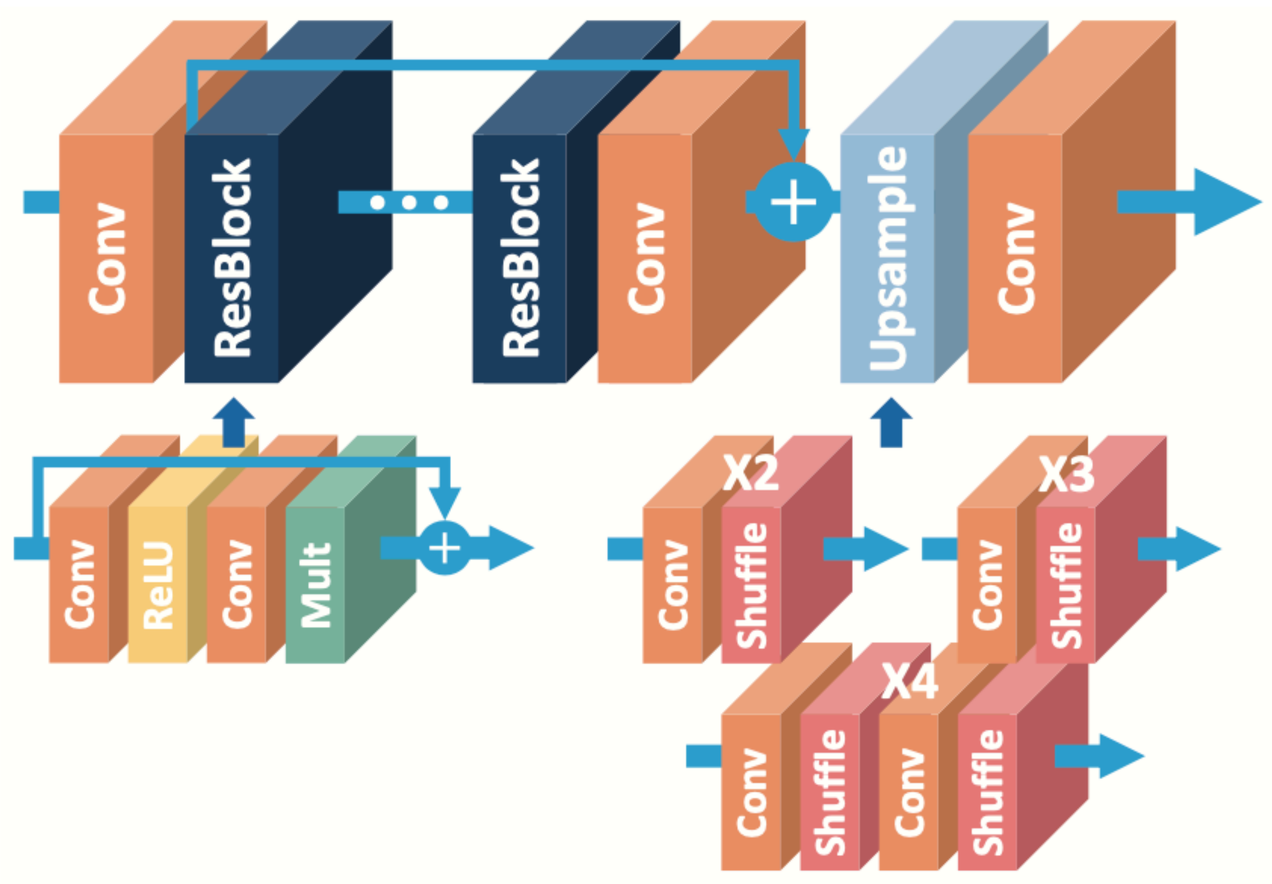


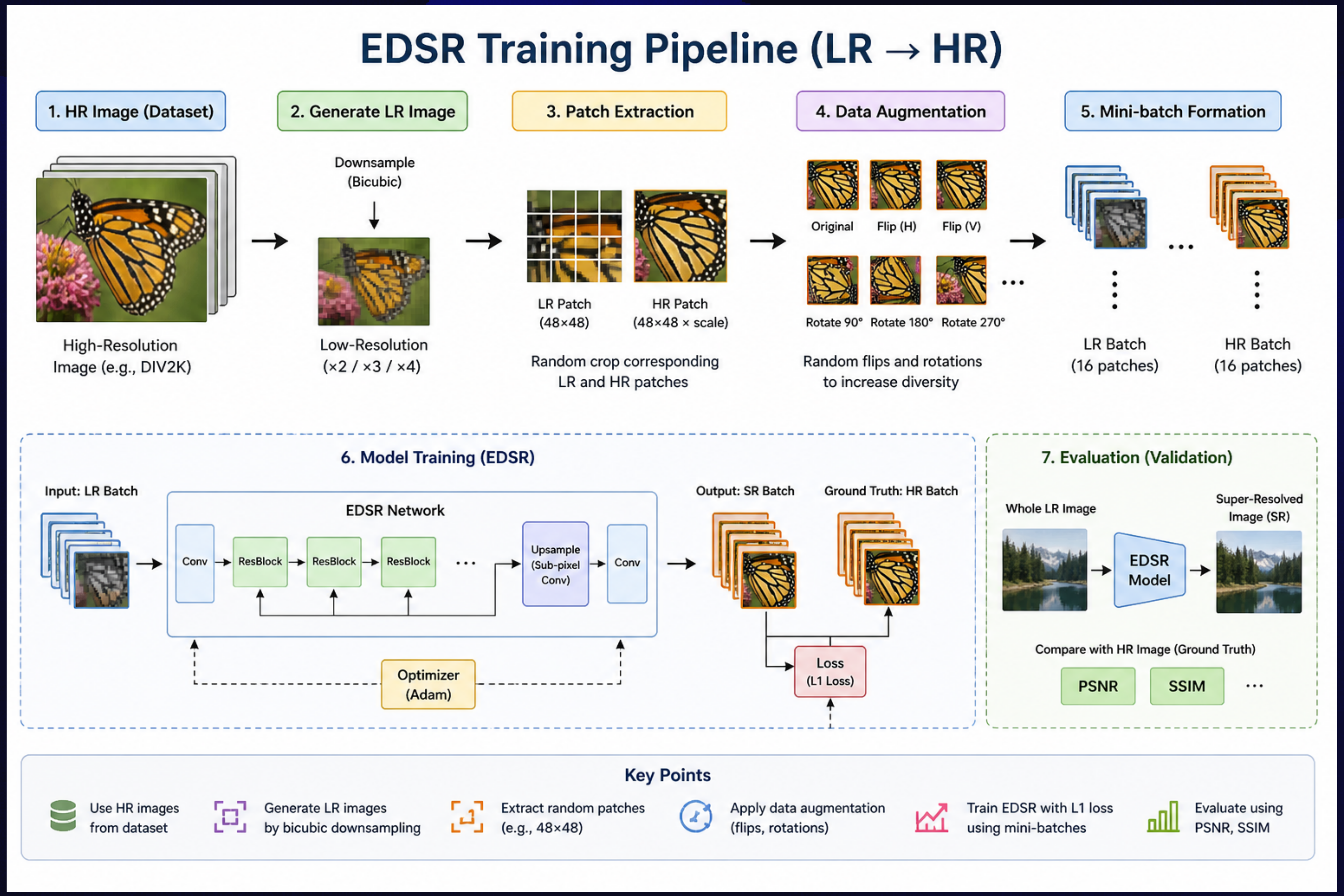
Figure 3: The architecture of the proposed single-scale SR network (EDSR).

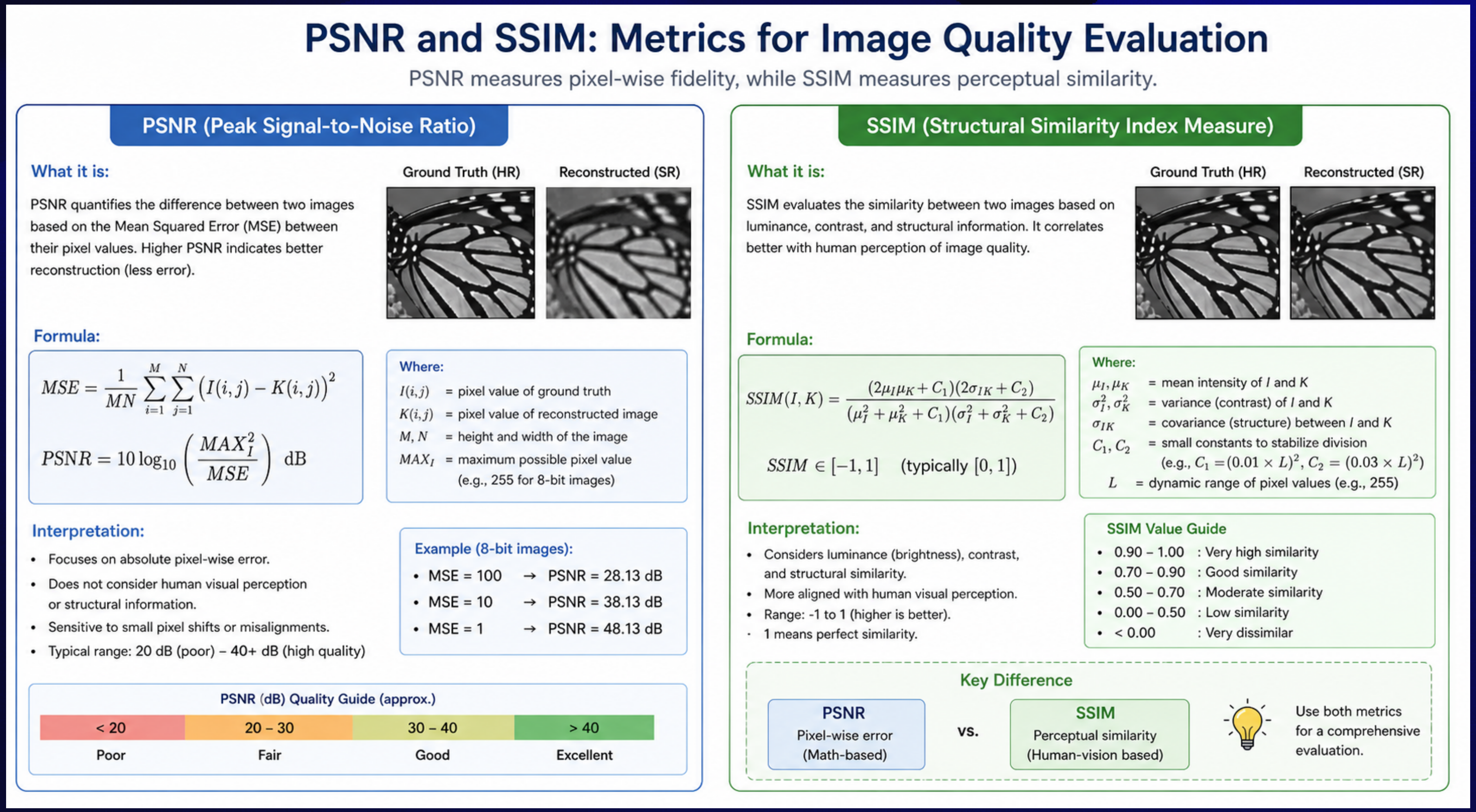
Options	SRResNet [14] (reproduced)	Baseline (Single / Multi)	EDSR	MDSR
# Residual blocks	16	16	32	80
# Filters	64	64	256	64
# Parameters	1.5M	1.5M / 3.2M	43M	8.0M
Residual scaling	-	-	0.1	-
Use BN	Yes	No	No	No
Loss function	L2	L1	L1	L1

Table 1: Model specifications.

Training Pipeline

- The training pipeline involves generating low-resolution (LR) images by downsampling high-resolution (HR) images, and then learning a mapping from LR to HR. Instead of using full images, random patches (e.g. 48 x 48) are extracted from LR images along with their corresponding HR patches to reduce memory usage and increase data diversity. Data augmentation such as random flips and rotations is applied to improve generalization. The model is trained in mini-batches (size 16) using L1 loss, which encourages sharper reconstructions, and optimized using gradient descent over many iterations. During validation, the full LR image is passed through the network and compared with the ground-truth HR image using metrics like PSNR and SSIM.

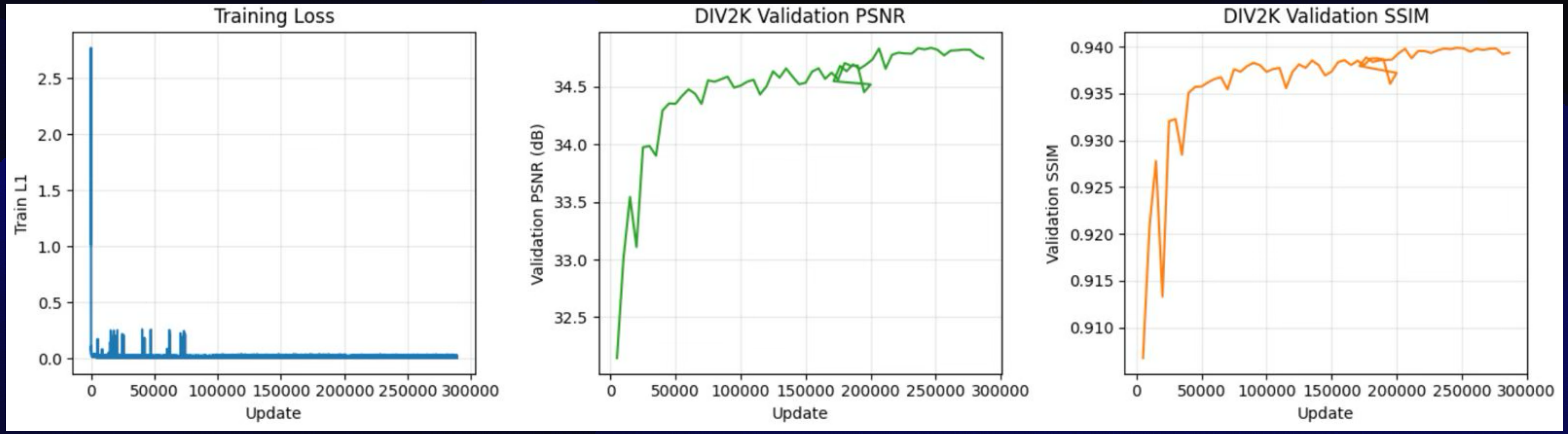




Results

- We have trained the model for a total of 290k iterations (where 1 iteration corresponds to 1 forward and backward pass of the whole model with batch (=16 here) no. of examples processed in parallel).

- The Training Loss (L1 Loss) and PSNR & SSIM scores on Validation dataset (equal to 100 images) are presented below

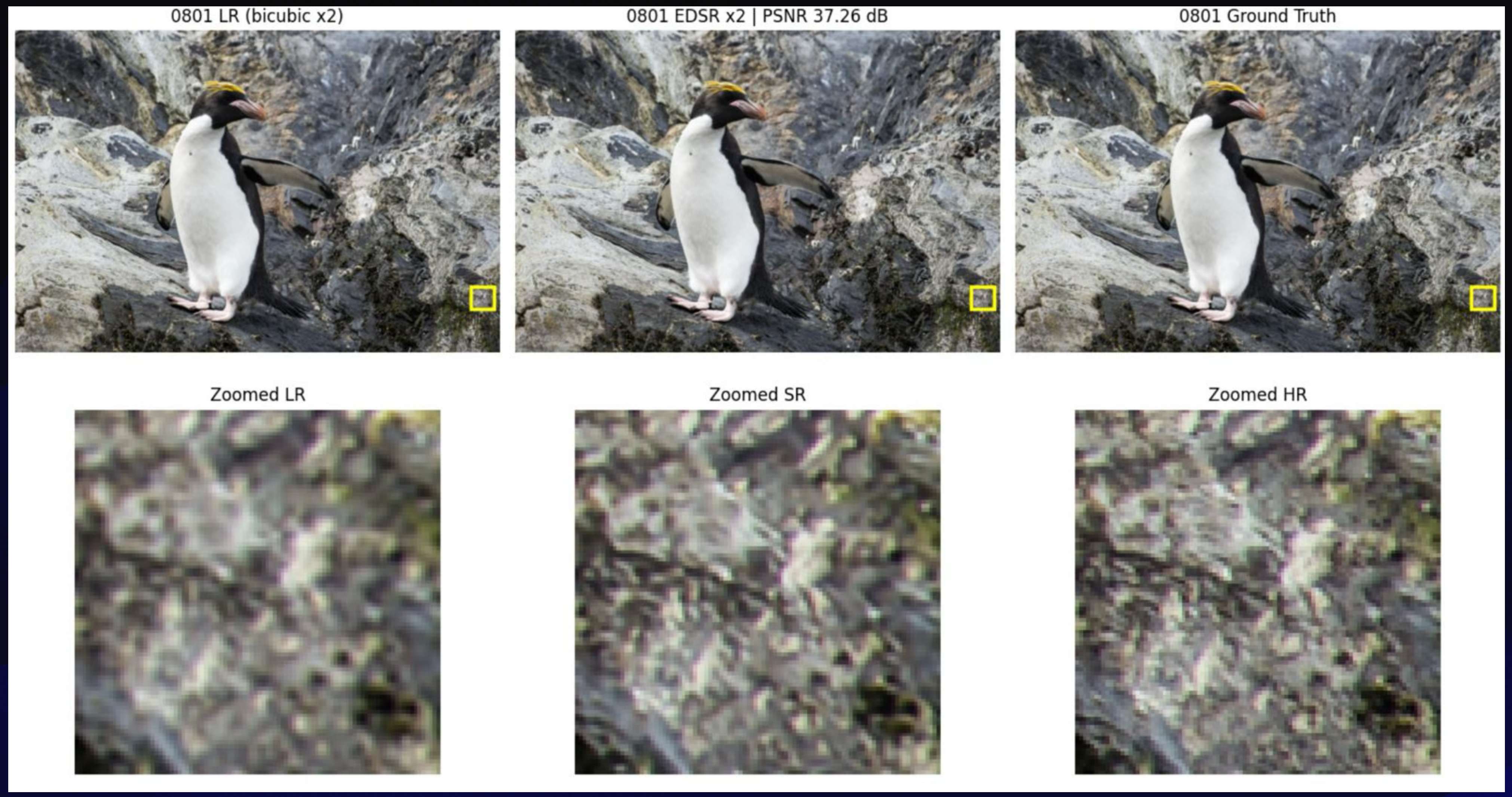


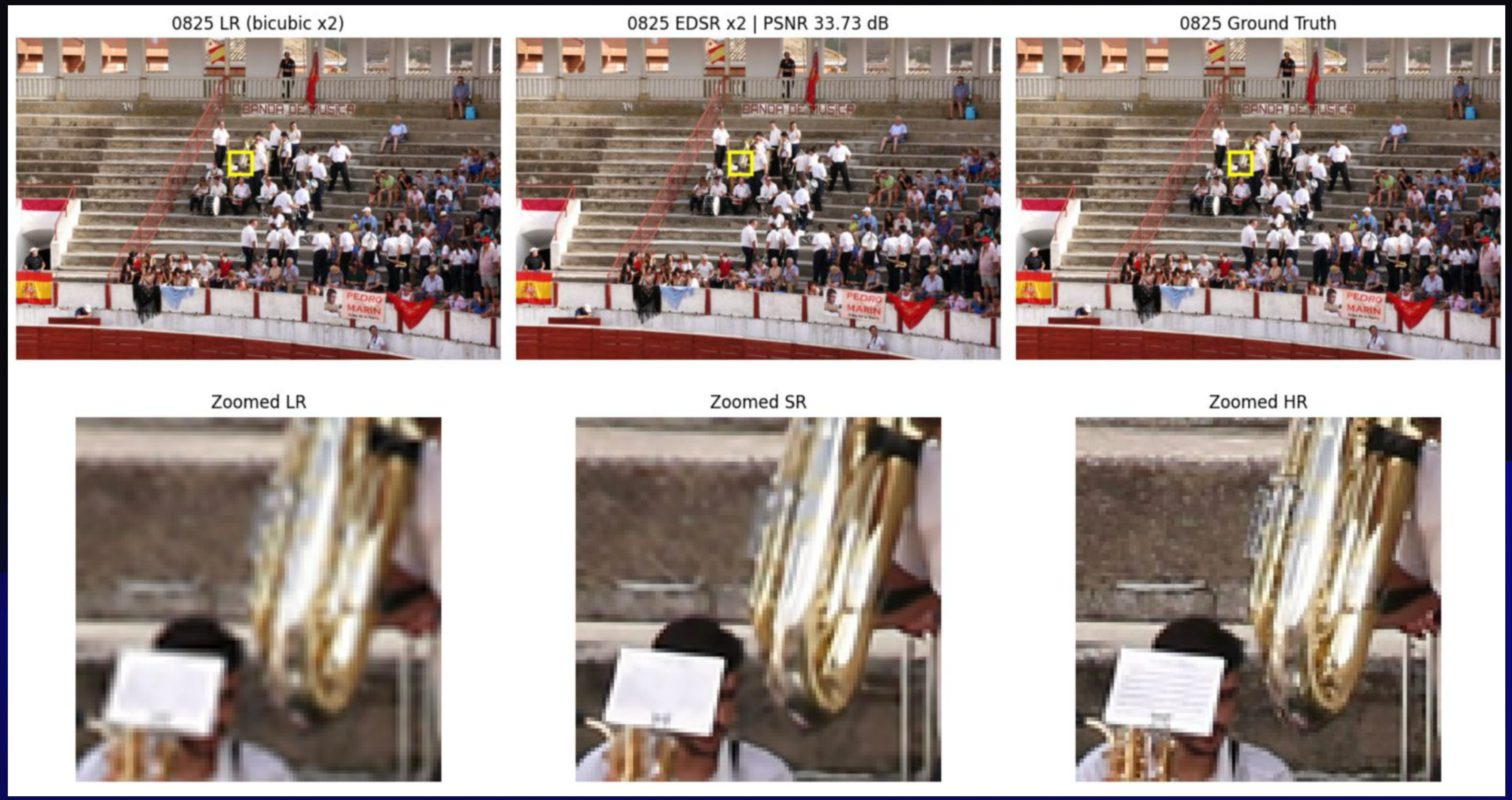

```
Loaded best checkpoint from update 235,000

======================================================================
DIV2K VALIDATION RESULTS (RGB, shave = 6 + scale)
======================================================================
Metric                        Bicubic          EDSR x2
----------------------------------------------------------------------
PSNR (dB)                     31.2523          34.8424
SSIM                          0.89808          0.93988
L1                                  -         0.012348
======================================================================
```

Results (Continued)

- The LR image, SR image generated by our model, and the corresponding true HR image are presented below





Conclusion

- The model implemented, meaningfully and successfully increases the resolution of an image with reasonably good PSNR & SSIM scores
- A few points to improve the implementation are listed below :-
- -> We have just trained over 290k iterations, we feel that it would be better if we can train the model over around 500k iterations, as from the graph itself, we can observe that we have not converged very well to a good score, and training for a bit longer (and choosing the best one till now), can be beneficial
- -> While downsampling the HR images to create LR images (that will be used as train inputs), we only use bicubic interpolation to downsample it, while it would be better if we used different ways / a combination of different ways like gaussian blur, others. This will help in generalizing the model to increase the resolution of an image in a better way (as it knows how to remove different types of “blurriness”
- -> We are not exactly sure how to implement it, but there are neural networks specifically made such that when you input data, they return you meaningful features, so we can try to use such a neural network (eg VGG) as a loss function to qualitatively replace the loss functions currently used

Thank You